

Hesam Amiri

CLOUD · EDGE · IOT
RESEARCH

Contact

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LOCATION

Bologna, Italy

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Languages

Persian *native* C2

English *upper-intermediate* B2

Italian *elementary* A2

German *elementary* A2

Technical skills

CLOUD & VIRTUALIZATION

OpenStack (Keystone, Nova, Neutron, Glance), Kubernetes (k0s, ConfigMap, Deployment, Service), Docker, cloud-init, Hetzner Cloud VPS

IOT & EMBEDDED

Arduino, NRF24L01 / NRF52, IEEE 802.15.4 (non-beacon), PCB design (Altium Designer), UART / ADC, sensor integration

NETWORKING & PROTOCOLS

TCP/IP, VXLAN, GRE, WireGuard, NETCONF/YANG, DNS, Linux netns, iptables/UFW, Wireshark

PROGRAMMING

Python (Flask, ncclient, asyncio), Bash, C, MATLAB, REST API design

AI & SIGNAL PROCESSING

Reinforcement learning (on-device, IEEE 802.15.4), supervised machine learning, signal processing, 5G systems, microwave engineering

Certifications

Supervised Machine Learning
DeepLearning.AI / Stanford · 2024

Industrial Automation — HMI & WinCC
PowerEn

MATLAB Certified
University of Birjand

IELTS Academic — Overall 6.5

MSc candidate in Telecommunications Engineering at the University of Bologna, with active research experience at WiLab applying reinforcement learning to MAC-protocol design on physical IoT hardware. Research interests span **cloud-native systems and edge computing** — particularly the orchestration of distributed services across the cloud-edge continuum, and the networking primitives (NETCONF, overlay tunnels, container runtimes) that enable them. Seeking an Erasmus+ traineeship in a research group working at this intersection.

— Research experience

• **RL-based dynamic MAC protocol for IoT networks** *2024 — present*

WiLab — Wireless Communication Laboratory, UniBO · Supervisor: Prof. Chiara Buratti

Ongoing research project replacing a static CSMA MAC protocol with a reinforcement-learning-driven dynamic alternative. Nodes learn transmission strategies adaptively against live wireless-channel conditions, without fixed scheduling rules. Validation conducted on a real IEEE 802.15.4 hardware testbed rather than in simulation — exposing the RL agent to genuine channel variability and hardware-level constraints. Implementation in Python; project supervised within the WiLab research group.

METHODS — Reinforcement learning, CSMA/CA, IEEE 802.15.4 hardware testbed, Python, on-device evaluation

— Research interests

Distributed systems and orchestration across the cloud-edge continuum. Network automation (NETCONF/YANG, declarative configuration) applied to softwarised infrastructure.

Reinforcement learning for adaptive resource management in wireless and edge environments. Container-native deployment models (Kubernetes, k0s) for latency-sensitive IoT workloads. Tunnel and overlay technologies for connecting distributed edge nodes to centralised cloud control planes.

— Education

• **M.Sc. Telecommunications Engineering** *2022 — 2027*

Alma Mater Studiorum — Università di Bologna · Italy

Core areas: Wireless Sensor Networks, 5G Systems, Microwave Engineering, Network Architecture, Distributed Systems, Network Automation. Currently enrolled, expected graduation 2027. Current weighted average: **25/30**.

• **B.Eng. Electrical & Electronics Engineering** *2019 — 2022*

University of Birjand · Iran

— Selected technical projects

• **OpenStack VM orchestration** *Apr 2026*

Distributed Systems M (Module 2), Prof. Davide Borsatti · UniBO

End-to-end Python automation against a multi-tenant OpenStack lab: token-based authentication via Keystone, external-network lookup, private network and subnet creation, router attachment with gateway, image and flavor selection, VM instantiation with cloud-init bootstrapping, and optional floating-IP assignment. Built as a

repeatable script for isolated tenant deployments — the same pattern that underpins production cloud-IoT backends.

STACK — Python, OpenStack SDK (Keystone, Neutron, Nova, Glance), cloud-init, OpenStack CLI

CODE — github.com/hesamAmiri04/distributed-systems-unibo/openstack

- **Kubernetes deployment of a containerised REST service** Apr 2026

Distributed Systems M (Module 2), Prof. Davide Borsatti · UniBO

Containerised a Python Flask REST server, exported and imported the image into a k0s cluster's containerd runtime, and deployed it through a complete manifest set: ConfigMap for runtime configuration injection, Deployment for pod lifecycle, NodePort Service for external access. Verified that the application loaded its configuration from the injected JSON file and that endpoints were reachable through the node IP. The same architecture used to expose IoT-ingest services in cloud-native deployments.

STACK — Docker, k0s, containerd, kubect1, Kubernetes (ConfigMap, Deployment, Service), Flask

CODE — github.com/hesamAmiri04/distributed-systems-unibo/kubernetes-introduction

- **Self-managed cloud VPN infrastructure** 2024 — present

Personal infrastructure project · Hetzner Cloud, Nuremberg

Self-directed systems-engineering project: provisioned an Ubuntu 22.04 VPS on Hetzner Cloud and built a working VPN service from the ground up using the VLESS+Reality protocol behind the Marzban control panel. Hands-on responsibility for the full server lifecycle — SSH hardening, firewall design with iptables / UFW, systemd unit management, DNS configuration, and ongoing operational reliability. The project exists outside any course requirement.

STACK — Ubuntu Server 22.04, Hetzner Cloud, VLESS+Reality, Marzban, iptables / UFW, systemd, SSH

- **Edge-IoT prototype — wireless sensor network** 2024

Wireless Sensor Networks course · UniBO

End-to-end low-power edge-IoT prototype for greenhouse monitoring. Edge node samples ambient light through an LDR sensor over ADC on Arduino Uno, transmits data over IEEE 802.15.4 in non-beacon mode, and forwards through a UART bridge to a coordinator node — the canonical edge-to-gateway-to-cloud pattern. Designed for extension toward cloud-side ingestion, which is precisely the architecture targeted by edge-computing research in distributed sensing.

STACK — Arduino, LDR sensors, ADC, UART, IEEE 802.15.4 (non-beacon), C

- **Custom PCB for NRF wireless IoT modules** 2023

Academic project · Altium Designer Enterprise

Designed a production-ready PCB for NRF24L01 and NRF52 wireless modules optimised for low-power IoT applications. Completed component placement, signal-integrity review and full DRC validation in Altium Designer Enterprise.

STACK — Altium Designer Enterprise, NRF24L01, NRF52, schematic capture, DRC

- **Additional projects**

Distributed Systems M and Network Automation, Prof. Davide Borsatti — UniBO, 2025/26

NETCONF / YANG automation — Python orchestrator using ncclient to configure three simulated network devices (RAN, aggregation, core) over NETCONF/SSH; atomic commits across the candidate datastore. [code]

RESTful API + Docker — Flask service with full CRUD and a task-execution endpoint, packaged into a reproducible Docker image (port 5001). [code]

Overlay-tunnel benchmarking — Comparative GRE / VXLAN / WireGuard tunnels between Linux netns; measured RTT and throughput (baseline 9.4 Gbps; WireGuard 7.6 Gbps). [code]